Perverse Sheaves

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1 Introduction

1.1 Overview of the Course

Given a topological space X, we can compute its singular cohomology $H^\bullet(X;k)$ over a commutative ring k.

Example 1.1.

$$H^i(\{\mathrm{pt}\};k) = \begin{cases} k, & (i=0), \\ 0, & (\textit{ otherwise }). \end{cases}$$

Example 1.2.

$$H^i(S^1;k) = \begin{cases} k, & (i=0,1), \\ 0, & (\textit{ otherwise }). \end{cases}$$

Definition 1.1. *Let X be a topological space. The i-th Betti number of X is*

$$b_i := \dim_{\mathbb{Q}} H^i(X;\mathbb{Q}).$$

Example 1.3. *In Example 1.2, the Betti numbers are*

$$b_i(S^1) = \begin{cases} 1, & (i=0,1), \\ 0, & (\textit{ otherwise }). \end{cases}$$

Theorem 1.1 (Poincaré Duality)**.** *Let M be a compact oriented manifold without boundary of dimension d, then for any $0 \le i \le d$, we have*

$$H^i(M;\mathbb{Q}) \cong H^{d-i}(M;\mathbb{Q})^\vee.$$

In terms of Betti numbers, this gives us

$$b_i(M) = b_{d-i}(M).$$

Theorem 1.2 (Hard Lefschetz)**.** *X a smooth complex projective algebraic variety of dimension $\dim_{\mathbb{C}} X = d$. Then there is an isomorphism*

$$H^i(X,\mathbb{Q}) \to H^{2d-i}(X,\mathbb{Q})$$

given by multiplication with some power of $l \in H^2(X,\mathbb{Q})$.

$$H^i(X,\mathbb{Q}) \xrightarrow{\ l\ } H^{i+2}(X,\mathbb{Q}) \xrightarrow{\ l\ } \cdots \xrightarrow{\ l\ } H^{2d-i}(X,\mathbb{Q})$$
$$\sim$$

We then get $b_i \le b_{i+2}$ if $i < d$ and $b_i \ge b_{i+2}$ if $i \le d$.

$$b_0 \le b_2 \le \cdots \le b_{\mathrm{middle}} \ge \cdots \ge b_{2d},$$
$$b_1 \le b_3 \le \cdots \le b_{\mathrm{middle}'} \ge \cdots \ge b_{2d-1}.$$

Definition 1.2. *For a complex projective variety X, we define*

$$H^{ij}(X) := H^i(X, \Omega_X^j).$$

Theorem 1.3.

$$H^k(X, \mathbb{C}) = \bigoplus_{i+j=k} H^{i,j}(X),\ H^{i,j} \cong \overline{H^{j,i}(X)}.$$

Furthermore,

$$b_k = \sum_{i+k} b_{i,j}, b_{i,j} = b_{j,i}.$$

Corollary 1.1. *In the situation of Theorem 1.4, we have k is odd $\Rightarrow$ b_k is even.*

Notation 1.1. *For a topologica space X, the constant sheaf of an abelian group A is denoted by $\underline{A}_X$.*

Theorem 1.4 (Deligne)**.** *For be a smooth morphism $f : X \to Y$ between smooth projective complex varieties, we have*

$$Rf_*\underline{\mathbb{Q}}_X \cong \bigoplus_{i\in\mathbb{Z}} \mathcal{H}^i(Rf_*\underline{\mathbb{Q}}_X)[-i].$$

.

Remark 1.1. *Consider,*

$$S^1 \xrightarrow{f \text{ a Hopf map}} S^3 \to S^2$$

Then $Rf_*\underline{\mathbb{Q}}_{S^3}$. However, $\mathcal{H}^0(Rf_*\underline{\mathbb{Q}}_{S^3}) = \underline{\mathbb{Q}}_{S^2}$ and $\mathcal{H}^1(Rf_*\underline{\mathbb{Q}}_{S^3}) = \mathbb{Q}_{S^2}$. Therefore,

$$Rf_*\underline{\mathbb{Q}}_{S^3} \ncong \underline{\mathbb{Q}}_{S^2} \oplus \underline{\mathbb{Q}}_{S^2}[-1], H^*(S^3) \ncong H^*(S^2) \oplus H^{*-1}(S^2).$$

Hence f does not satisfy Theorem 1.4.

Question. *Over which space, does the Poincaré duality hold?*

This question led to the invention of the following notion.

Definition 1.3 (Intersection Cohomology)**.** *For a variety X over $\mathbb{C}$, we associate an mathematical object $\mathrm{IH}^\bullet(X; \mathbb{Q})$ which is called intersection cohomology.*

Theorem 1.5. *For a proper variety X over $\mathbb{C}$ of finite dimension, we have*

$$\mathrm{IH}^i(X; \mathbb{Q}) \cong \mathrm{IH}^{2d-i}(X; \mathbb{Q})^\vee.$$

Theorem 1.5 proves some versions of Theorem 1.2.

Notation 1.2. *Let X be a topological space. The derived category of constructible sheaves on X is denoted by $D_c^b(X)$.*

Theorem 1.6. *For a proper variety X over $\mathbb{C}$ of finite dimension, there is a complex of sheaves $\mathrm{IC}_X \in D_c^b(X)$ such that*

$$\mathrm{IH}^*(X, \mathbb{Q}) \cong H^*(X, \mathrm{IC}_X).$$

Definition 1.4. *Such IC_X in Theorem 1.6 is called a perverse sheaf.*

Remark 1.2. *Perverse sheaves were introduced by Belinson-Bernstein-Deligne in 1983. Together with Gabber, they extended Theorem 1.4 to the case when $f : X \to Y$ is proper.*

1.2 Preliminaries

Notation 1.3. *Let X be a topological space and k be a commutative ring. The category of presheaves of k-modules over X is denoted by $\mathrm{PreShv}(X;k)$, similarly the sheaves of k-modules over X is denoted by $\mathrm{Shv}(X;k)$.*

Example 1.4. *For all $M \in \mathrm{Mod}(k)$, there is a constant sheaf $\underline{M}_X \in \mathrm{Shv}(X;k)$.*

Definition 1.5. *A local system on a topological space X is a sheaf on X which is locally a constant sheaf of an abelian group.*

Example 1.5. *For $r > 0$, consider morphisms of groups,*

$$p : \pi_1(X, x_0) \to \mathrm{Aut}(k^{\oplus r}).$$

Define L_p as follows. Let $\pi : \tilde{X} \to X$ be the universal covering space. For $U \subseteq X$,

$$L_p := \{\pi^{-1}(U) \to k \mid \forall g \in \pi_1(X, x_0), f(x \cdot g) = p(g)^{-1} f(x)\}.$$

Then L_p is a local system on X.

Theorem 1.7. *$\mathrm{Shv}(X;k)$ is an abelian category.*

Proposition 1.1. *The canonical inclusion $\mathrm{Shv}(X;k) \to \mathrm{PreShv}(X;k)$ admits a left adjoint and such functor is called the sheafification,*

$$\mathrm{PreShv}(X;k) \ni \mathcal{F} \mapsto \mathcal{F}^\dagger \in \mathrm{Shv}(X;k).$$

Explicitly, for $\mathcal{F} \in \mathrm{PreShv}(X;k)$ and $\mathcal{G} \in \mathrm{Shv}(X;k)$, we have

$$\mathrm{Mor}_{\mathrm{PreShv}}(\mathcal{F}, \mathcal{G}) \cong \mathrm{Mor}_{\mathrm{Shv}}(\mathcal{F}^\dagger, \mathcal{G}),$$

which is natural. In particular, setting $\mathcal{G} = \mathcal{F}^\dagger$, there is a natural map $\mathcal{F} \to \mathcal{F}^\dagger$ which is obtained from $\mathrm{id} \in \mathrm{Mor}_{\mathrm{Shv}}(\mathcal{F}^\dagger, \mathcal{F}^\dagger)$.

Proposition 1.2. *The natural morphism $\mathcal{F} \to \mathcal{F}^\dagger$ induced isomorphisms on stalks.*

Definition 1.6. *Given a complex of sheaves,*

$$\mathcal{F}^\bullet := \cdots \to \mathcal{F}^{i-1} \overset{d^{i-1}}{\to} \mathcal{F}^i \overset{d^i}{\to} \mathcal{F}^{i+1}.$$

we define its cohomology of sheaves to be

$$\mathcal{H}(\mathcal{F}^\bullet) := \ker d^i \oplus \mathrm{im}\, d^{i-1}.$$

A morphism of complexes is called a quasi-isomorphism if it induces isomorphisms in cohomology.

Definition 1.7. *The derived category on $\mathrm{Shv}(X;k)$ is a category satisfying the following.*

i. The objects are complex of sheaves,

$$\mathcal{F}^\bullet := \cdots \to \mathcal{F}^{i-1} \overset{d^{i-1}}{\to} \mathcal{F}^i \overset{d^i}{\to} \mathcal{F}^{i+1}.$$

ii. The morphisms are morphisms of complexes.

iii. Quasi-isomorphisms are invertible.

Remark 1.3. *The derived category of sheaves on X over k is a triangulated category.*

However the derived category turns out to be to large to handle. Therefore, we will use a smaller category called the bounded constructible derived category of sheaves denoted by $D^b_c(X;k)$.

Remark 1.4. *The category of bounded constrictible derived category of sheaves $D^b_c(X)$ is a subcategory of the derived category of sheaves which is more manageable.*

Definition 1.8. *There exists a stratification in locally closed subspaces $X = \bigsqcup_\alpha S_\alpha$. That is for $\mathcal{F}^\bullet \in D_c^b(X)$, we have $\mathcal{H}^i(\mathcal{F}^\bullet)|_{S_\alpha}$ is a finite rank local system and $\mathcal{H}^i(\mathcal{F}^\bullet) = 0$ for $|i| \gg 0$.*

Remark 1.5. *Let $f : X \to Y$ be a continuous map. The following so-called six-functors are important when it comes to produce example of sheaves,*

$$Rf_* \dashv f^*, f_! \dashv f^!, (\cdot \otimes \cdot) \dashv \mathrm{Hom}(\cdot, \cdot).$$

Remark 1.6. *If f is proper then $Rf_* = f_!$, if f is open immersion then $f^* = f^!$.*

Theorem 1.8.

$$D_c^b(X; k) \cong D^b(\mathrm{Shv}_c(X; k)).$$

Example 1.6. *For a scheme X and $\alpha : X \to \mathrm{pt}$, we have,*

$$\alpha_* \underline{k}_{\mathrm{pt}} \cong \underline{k}_X, \alpha^! \underline{k}_{\mathrm{pt}} \cong \omega_X.$$

With the aid of these functors, we define(reconstruct) following complexes.

Definition 1.9.

$$\begin{aligned} H^\bullet(X; k) &:= \alpha_* \alpha^* \underline{k}_{\mathrm{pt}}, \\ H^{\mathrm{BM}}(X; k) &:= \alpha_* \alpha^! \underline{k}_{\mathrm{pt}}, \\ H_c^*(X; k) &:= \alpha_! \alpha^* \underline{k}_{\mathrm{pt}}, \\ H_*(X; k) &:= \alpha_! \alpha^! \underline{k}_{\mathrm{pt}}. \end{aligned}$$

Example 1.7. *Let $Z \subseteq X$ be closed and $U = X \backslash Z$. Consider $Z \overset{i \ closed}{\hookrightarrow} X \overset{j \ open}{\hookleftarrow} U$. This gives natural transformations,*

$$(\cdot) \to i_* i^*(\cdot), (\cdot) \to j_* j^*(\cdot), (\cdot) \to i_! i^!(\cdot), (\cdot) \to j_! j^!(\cdot).$$

Furthermore, this gives an exact triangle,

$$j_! j^!(\cdot) \to (\cdot) \to i_* i^*(\cdot) \to [+1].$$

Using Remark 1.6 and plugging in $\underline{k}_X = \alpha_* \underline{k}_{\mathrm{pt}}$, we obtain

$$\alpha_* j_! \underline{k}_U \to \alpha_* \underline{k}_X \to \alpha_* i_* \underline{k}_Z \to$$

Suppose $X = U \cup V$ and $W = U \cap V$, and there are natural transformations,

$$(\cdot) \to (j_V)_* (j_U)^*(\cdot).$$

In the category of sheaves $\mathrm{Shv}(X; k) \subseteq D_c^b(X; k)$.

$$\begin{array}{ccc} \mathrm{Shv}(X; k) & \hookrightarrow & D^b(\mathrm{Shv}(X; k)) \\ \uparrow & & \uparrow \\ \mathrm{Shv}_c(X; k) & \hookrightarrow & D_c^b(\mathrm{Shv}(X; k)) \end{array}$$

Definition 1.10. *The Perverse sheves is $\mathrm{Perv}(X; k) \subseteq D_c^b(\mathrm{Shv}(X; k))$, where X is fnite-type variety over a field.*

Explicitly, for $D(\cdot) = \mathcal{H}om(\cdot, \omega_X)$, define $\mathcal{F} \in D_c^b(X; k)$ is a perverse sheaf if for all $n \in \mathbb{Z}$, $\dim \mathrm{supp}\, \mathcal{H}(\mathcal{F}) \leq i$ and $\dim \mathrm{supp}\, \mathcal{H}(D(\mathcal{F})) \leq i$.

Example 1.8. *Let X be a smooth variety over $\mathbb{C}$ of dimension $\dim_{\mathbb{C}} X = d$, then,*

$$\omega_X = k_X[2d],$$

then $\underline{k}_X[d]$ is a perverse sheaf and $\mathbb{D}(\underline{k}_k[d]) = \underline{k}_X[d]$. As,

$$\mathcal{H}(k_x[d]) = \begin{cases} \underline{k}_X & (i = d), \\ 0, & (\ otherwise\ ). \end{cases}$$

Hence

$$\dim \operatorname{supp}(\mathcal{H}(\underline{k}_X[d])) = \begin{cases} d, & (i = d), \\ -\infty, & (\ otherwise\ ). \end{cases}$$

Similarly, if Lis a local system on X, then $L[d] \in \operatorname{Perv}(X; k)$.

Theorem 1.9. $\operatorname{Perv}(X; k)$ *is an Abelian category which is both Noetherian and Artinian.*

Every perverse sheaf has a composition series with ? All simple objects are constructed as follows,

$$\underbrace{X}_{\text{smooth finite-type variety}} \hookleftarrow \underbrace{Z}_{\text{closed subvariety}} \hookleftarrow \underbrace{U}_{\text{open smooth}}.$$

Let L be a local system on V, then we construct a perverse sheave,

$$\operatorname{IC}_Z(L) \subseteq \operatorname{Perv}(X; k),$$

If it is supported on Z and

$$\operatorname{IC}_Z(L)|_V = L[\dim V].$$

References

[1] Pramod Achar, Perverse Sheaves and Applications to Representation Theory.

[2] Mark de Cataldo, Luca Migliorini, The Decomposition Theorem and the topology of algebraic maps.